Week 6
Exception Handling and Multithreading
Program 16
Write a Java program demonstrating try, catch, finally, throw, and throws.

```
class ExceptionDemo {

   static void checkNumber(int n) throws Exception {
      if (n < 0) {
         throw new Exception("Negative number is not allowed");
      }
      System.out.println("Number: " + n);
   }

   public static void main(String[] args) {
      try {
         int a = 10 / 0;
         System.out.println(a);
      } catch (ArithmeticException e) {
         System.out.println("Cannot divide by zero");
      } finally {
         System.out.println("Finally block executed");
      }

      try {
         checkNumber(-5);
      } catch (Exception e) {
         System.out.println(e.getMessage());
      }
   }
}
```

OUTPUT:
Cannot divide by zero
Finally block executed
Negative number is not allowed

Program 17
Write a Java program to create a user-defined exception for invalid age.

```
class InvalidAgeException extends Exception {
   InvalidAgeException(String message) {
      super(message);
   }
}

class AgeCheck {
   static void checkAge(int age) throws InvalidAgeException {
      if (age < 18) {
         throw new InvalidAgeException("Invalid age: Not eligible to vote");
      }
      System.out.println("Eligible to vote");
   }

   public static void main(String[] args) {
      try {
         checkAge(16);
```

```
      } catch (InvalidAgeException e) {
         System.out.println(e.getMessage());
      }
   }
}
```

OUTPUT:
Invalid age: Not eligible to vote

Program 18
Write a Java program to create multiple threads using Thread class and Runnable interface.

```
class MyThread extends Thread {
   public void run() {
      for (int i = 1; i <= 3; i++) {
         System.out.println("Thread Class: " + i);
      }
   }
}

class MyRunnable implements Runnable {
   public void run() {
      for (int i = 1; i <= 3; i++) {
         System.out.println("Runnable Interface: " + i);
      }
   }
}

class ThreadDemo {
   public static void main(String[] args) {
      MyThread t1 = new MyThread();

      MyRunnable r = new MyRunnable();
      Thread t2 = new Thread(r);

      t1.start();
      t2.start();
   }
}
```

OUTPUT:
Thread Class: 1
Thread Class: 2
Thread Class: 3
Runnable Interface: 1
Runnable Interface: 2
Runnable Interface: 3